

Deliverable 1 — Problem Formulation

Introduction to Machine Learning

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1 Context and Motivation

With the exponential growth of digital traffic and the increasing sophistication of cyberattacks, the efficient detection of malicious behavior has become a critical challenge for organizations and digital infrastructures. Traditional approaches based on predefined rules and signatures present significant limitations, particularly in detecting new or previously unknown attacks (zero-day attacks).

In this context, Machine Learning techniques, especially anomaly detection methods, emerge as a promising solution. These approaches allow modeling normal network behavior and identifying deviations that may indicate malicious activity, without relying solely on previously labeled attack data.

This project proposes the development of a machine learning system for anomaly detection in network traffic, aiming to automatically identify suspicious patterns that may correspond to cyberattacks.

2 Dataset Description

The dataset selected for this project is the **CIC-IDS2017**, developed by the Canadian Institute for Cybersecurity.

This dataset contains realistic network traffic data, including both benign activity and multiple types of attacks, such as:

- Distributed Denial of Service (DDoS)
- Brute Force
- Port Scanning
- Web Attacks

It includes a wide range of statistical features extracted from network flows, such as connection duration, number of packets, transmitted bytes, and temporal metrics.

2.1 Limitations and Potential Bias

- The dataset was generated in a controlled environment and may not fully represent real-world scenarios
- Certain attack types may be overrepresented
- Possible lack of up-to-date attack patterns
- Dependence on the quality and selection of provided features

3 Type of Machine Learning Problem

The problem will be addressed as an **anomaly detection task**, within an unsupervised or semi-supervised learning framework.

The model will be trained primarily using data representing normal network traffic, learning the typical behavior patterns of the system. Observations that significantly deviate from these learned patterns will be identified as anomalies.

This approach is particularly relevant in cybersecurity, as it enables the detection of previously unseen attacks without requiring labeled examples.

4 Features and Target Definition

4.1 Input Features

The model will use features extracted from network traffic, including:

- Connection duration
- Number of transmitted packets
- Volume of data (bytes sent/received)
- Data transfer rate
- Number of connections per time unit
- Network protocols used

These features allow capturing behavioral patterns of network activity and distinguishing between normal and anomalous traffic.

4.2 Target

In the context of anomaly detection, there is no explicit target variable during training. The model learns patterns from normal data and identifies deviations as anomalies.

However, the labeled data available in the dataset (benign vs attack) will be used for performance evaluation.

5 Evaluation Strategy

The model will be evaluated based on its ability to correctly identify anomalous behavior, using metrics suitable for cybersecurity contexts:

- **Precision** — proportion of detected anomalies that are actual attacks
- **Recall** — ability to detect existing attacks
- **F1-score** — balance between precision and recall

In this context, particular emphasis will be placed on recall, as failing to detect attacks (false negatives) may lead to severe consequences.

6 Initial Risks and Assumptions

6.1 Risks

- High rate of false positives, classifying normal traffic as anomalous
- Difficulty in detecting attacks that closely resemble normal behavior
- Potential overfitting to dataset-specific patterns
- Sensitivity to feature quality and selection

6.2 Assumptions

- The normal traffic in the dataset represents realistic behavior
- The available features are sufficient to distinguish normal and anomalous patterns
- The model will generalize to unseen data

7 Final Considerations

This project aims to explore the potential of anomaly detection techniques in cybersecurity, contributing to the development of more adaptive and robust systems for threat detection.

The proposed approach not only addresses the defined problem but also promotes critical reflection on the challenges associated with real-world attack detection, including data limitations, operational risks, and ethical implications.